EFY TESTED Ashwini K. Sinha

A REMOTE WITHOUT BATTERY That Can Work For Years

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There is huge chemical waste being generated by billions of batteries used in remote controllers for electronic devices like TV sets, set-top boxes, air-conditioners, and even some ceiling fans now-a-days. The waste produced by their spent batteries keeps increasing annually.

This DIY project harvests energy from the environment to power the remote controller and does not require any battery. Since low power is needed for a remote's circuit, the battery can be eliminated altogether and thus save tons of chemical waste each year. The author's prototype is shown in Fig. 1.

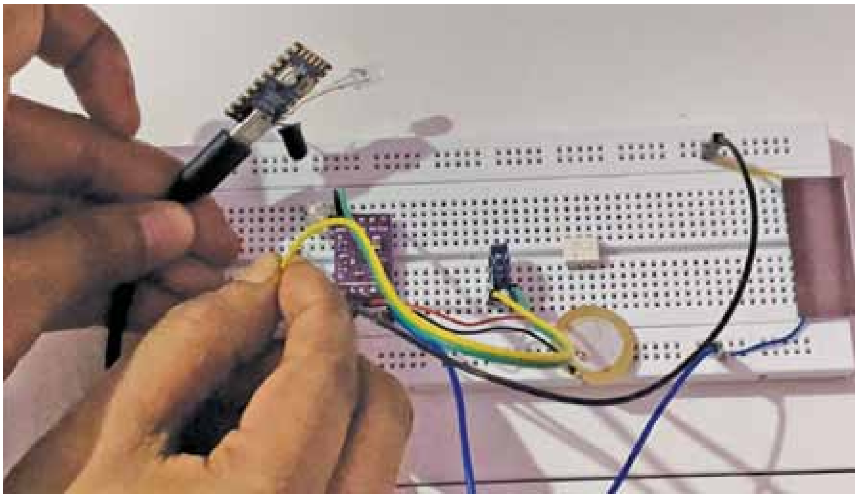

Fig. 1: Author's prototype on a breadboard

Circuit and working

The circuit diagram of this remote controller can be divided into two sections—the transmitter and the receiver. Circuit diagram of the transmitter section is shown in Fig. 2.

The transmitter is built around energy-harvesting module LTC3588 (MOD1), two 5.5V, 1F super capacitos (C1 and C2), small 2.2V, 50mA solar cell (SC1), piezoelectric crystal (PZ1), and transmitting LED (IR1). The piezoelectric crystal is connected to input pins PZ1 and PZ2 of LTC3588. The solar cell is connected to Vin and GND pins. At the output the IR LED is connected between D0 and the ground pins of MOD1.

Circuit diagram of the receiver is shown in Fig. 3, which also includes circuit diagram of the transmitter for convenience. It is built mainly around MOD2, receiving LED/ TSOP4538 (IR2), and 5mm LED

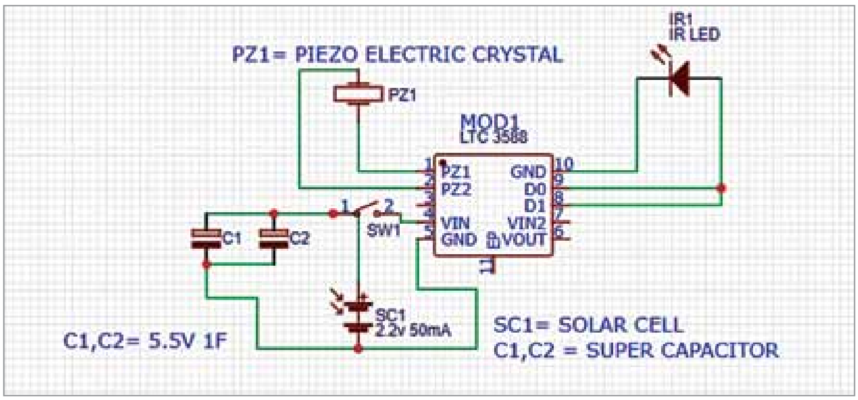

Fig. 2: Remote transmitter circuit

BILL OF MATERIAL

Component	Quantity	Description	Abbreviation
Solar Cell	1	2.2V, 50mA	SC1
Super Capacitor	2	5.5V, 1F	C1, C2
Piezoelectric Crystal	1	Piezoelectric generator	PZ1
IR LED	1	2.2V, 20mA	IR1
Energy Harvesting Module	1	LTC3588	MOD1
IR Receiver	1	TSOP4538/IR Receiver LED	IR2
RP2040 Zero Module	1	Microcontroller used for receiver testing	MOD2
5mm LED	1	LED to transmit signal	LED1